CHAPTER 4 APPLICATIONS OF DIFFERENTIATION

4.1 Maxima and Minima

Nature of graph } limit, differentiation

- **Definition**

Let D, the domain of f, contain the point c. We say that

(i) $f(c)$ is the **absolute maximum** value of f on D if $f(c) \geq f(x)$ for all x in D;

(ii) $f(c)$ is the **absolute minimum** value of f on D if $f(c) \leq f(x)$ for all x in D;

(iii) $f(c)$ is an absolute extreme value of f on D if it is either the absolute maximum value or the absolute minimum value;

- **Definition**

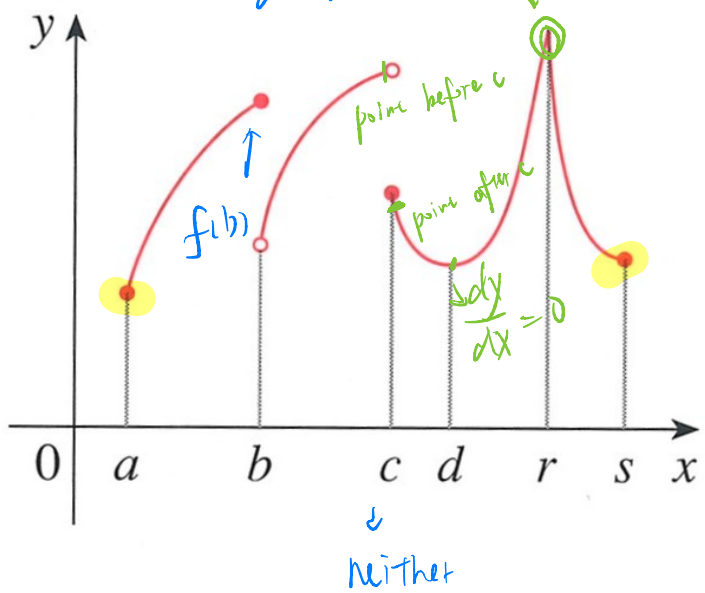
Let D, the domain of f, contain the point c. We say that

(i) $f(c)$ is a **local maximum** value of f if $f(c) \geq f(x)$ when x is near c;

(ii) $f(c)$ is a **local minimum** value of f if $f(c) \leq f(x)$ when x is near c;

(iii) $f(c)$ is a local extreme value of f on D if it is either a local maximum value or a local minimum value.

Example 4.1. Find the absolute extreme values and the local extreme values of the function.



- **Extreme Value Theorem**

> If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum value and an absolute minimum value there.

- If c is a point at which $f'(c) = 0$, we call c a **stationary point**.

Example 4.2. Find the stationary points of the function $f(x) = 2x^3 - 3x^2 + 2$.

$f'(c) = 0$

$f'(x) = 6x^2 - 6x$

when at stationary point, $f'(c) = 0$

$6x^2 - 6x = 0$

$6x(x-1) = 0$

$x = 0$ or $x = 1$

① The stationary point are at $x=0$ or at $x=1$.

② The stationary points are $(0, 2)$ and $(1, 1)$

① or ② ✓

- If c is an interior point of the domain of f such that $f'(c)$ does not exist, we call c a **singular point**.

c is defined but not differentiable

vertical tangent line / corner

Example 4.3. Find the singular points of the function $y = \sqrt[3]{x^2}$.

$y = x^{\frac{2}{3}}$

$y' = \frac{2}{3} x^{-\frac{1}{3}} = \frac{2}{3\sqrt[3]{x}}$

$\frac{dy}{dx}$ DNE when $x=0$, but x is an interior point of the domain

$\Rightarrow$ y has a singular point at $x=0$

- If f is a function defined on an interval, a **critical point** c of f is one of the following three types of points:

  (a) an end point of I;

  (b) a stationary point of f;

  (c) a singular point of f.

- **Fermat's Theorem**

> Let f be defined on an interval I containing the point c. If $f(c)$ is a local extreme value, and $f'(c)$ exists, then $f'(c) = 0$.

- **Critical Point Theorem**

> Let f be defined on an interval I containing the point c. If $f(c)$ is an extreme value, then c must be a critical point.

- In order to find the extreme values of a function f that is continuous on a closed interval $I = [a, b]$,

  Step 1 Find the critical points of f on I.

  Step 2 Evaluate f at each of these critical points. The largest value is the maximum value; the smallest value is the minimum value.

Example 4.4. Find the absolute maximum and absolute minimum values of the function $f(x) = x^3 - 3x^2 + 1$ on $\left[-\frac{1}{2}, 4\right]$.

① endpoints : $x = -\frac{1}{2}$, $x = 4$

$f(-\frac{1}{2}) = \frac{1}{8}$

$f(4) = 17$ (abs max)

② stationary point : $f'(x) = 0$

$3x^2 - 6x = 0$

$3x(x-2) = 0$

$x = 0$ or $x = 2$

$f(0) = 1$

$f(2) = -3$ (abs min)

③ Singular point : $f'(x) = 3x^2 - 6x$ always defined. thus no singular point

Absmax value is 17 , and absmin value is -3

Example 4.5. Find the absolute maximum and absolute minimum values of $F(x) = x^{\frac{2}{3}}$ on $[-1, 2]$. Conclusion: F(x) has a absmax of $\sqrt[3]{4}$, absmin of 0.

① endpoint : $x = -1$, $x = 2$

② stationary point . $F'(x) = \frac{2}{3} x^{-\frac{1}{3}} = \frac{2}{3\sqrt[3]{x}}$

$F'(x) \neq 0$ is impossible .

$F'(x)$ has no stationary point.

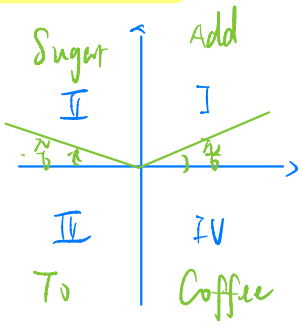
③ Singular point : $F'(x)$ DNE when $x = 0$, but $x = 0$ is a interior point of the domain. $F(x)$ has a singular point at $x = 0$

$F(0) = 0$ (absmin)

$F(-1) = 1$

$F(2) = \sqrt[3]{4} \approx 1.5874$ (absmax)

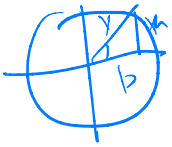
Example 4.6. Find the absolute maximum and absolute minimum values of $f(x) = x + 2\cos x$ on $[-\pi, 2\pi]$.



① endpoint : $x = -\pi$, $x = 2\pi$

② stationary point : $f'(x) = 1 - 2\sin x = 0$

$x = \sin^{-1}(\frac{1}{2})$



$\sin = \frac{a}{y} = a$

$\cos = \frac{b}{y} = b$

$x = \frac{\pi}{6}$ or $\frac{5\pi}{6}$

and $x = -\frac{\pi}{6}$ or $-\frac{5\pi}{6}$ sin 後要正負.

③ Singular point $f'(x) = 1 - 2\sin x$ is always defined $\forall x \in \mathbb{R}$.

Thus, $f(x)$ has no singular point.

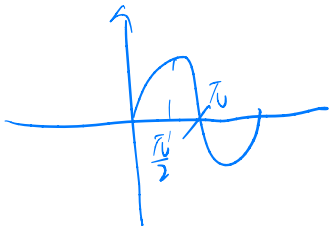
$f(x) = x + 2\cos x$

$f(-\pi) = -\pi - 2 = -5.1416$ (abs min)

$f(2\pi) = +2\pi + 2 = 8.2832$ (abs max)

$f(\frac{\pi}{6}) = 2.3566$

$f(\frac{5\pi}{6}) = 0.8859$



$f(x)$

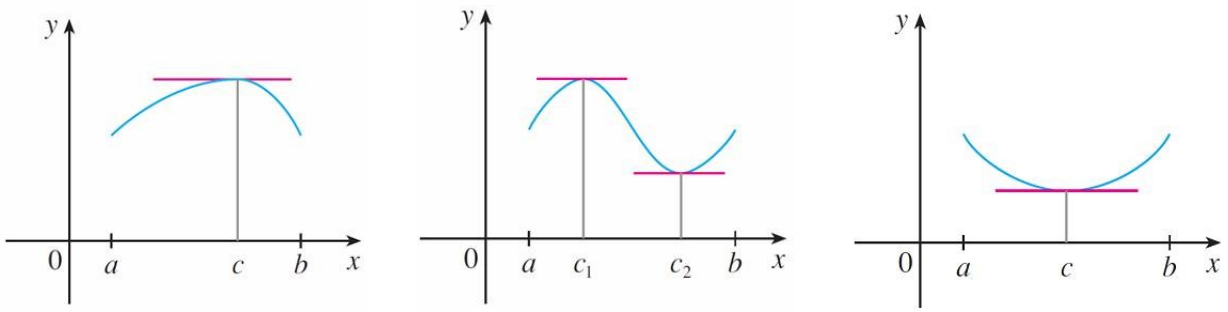
Exercises 4.1 5, 17, 27, 47, 53, 57.

4.2 The Mean Value Theorem for Derivatives

- **Rolle's Theorem**

If f is continuous on a closed interval $[a, b]$ and differentiable on its interior (a, b), and $f(a) = f(b)$, then there is a c in (a, b) where

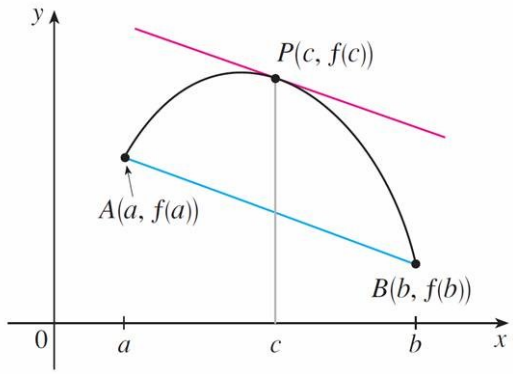
$$f'(c) = 0$$

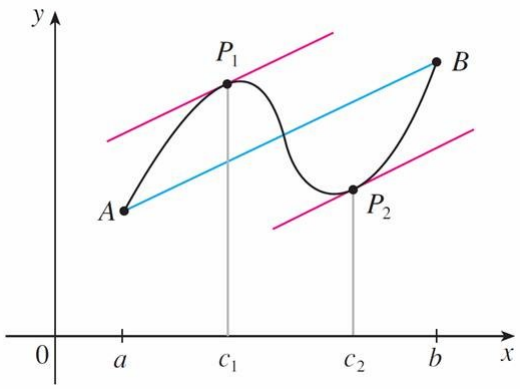


- **The Mean Value Theorem for Derivatives**

If f is continuous on a closed interval $[a, b]$ and differentiable on its interior (a, b), then there is a c in (a, b) where

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$





Example 4.7. Find the number c guaranteed by the Mean Value Theorem for $f(x) = 2\sqrt{x}$ on $[1, 4]$.

$f(x) = 2x^{\frac{1}{2}}$

$f'(x) = 2 \times \frac{1}{2} x^{-\frac{1}{2}} = x^{-\frac{1}{2}} = \frac{1}{\sqrt{x}}$

证明 $f(x)$ continuous 的方法
① 直接求出 $f'(x)$
② $f(x)$ 存在

$f'(x)$ exists for $x \in [1, 4] \Rightarrow f(x)$ continuous on $[1, 4]$

MVT tells us there exists a $x = c$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{f(4) - f(1)}{4 - 1}$$

$$\frac{1}{\sqrt{c}} = \frac{2\sqrt{4} - 2\sqrt{1}}{3} = \frac{2}{3}$$

$$\frac{1}{\sqrt{c}} = \frac{2}{3} \Rightarrow c = \frac{9}{4}$$

Homework

Example 4.8. Let $f(x) = x^3 - x^2 - x + 1$ on $[-1, 2]$. Find all numbers c satisfying the conclusion of the Mean Value Theorem.

more than 1 c

$\because f(x)$ is polynomial $\Rightarrow$ continuous on $\mathbb{R}$

$f'(x) = 3x^2 - 2x - 1$ exists for all x on $[-1, 2] \Rightarrow$ differentiable

$\therefore$ MVT tells us there exits a $x = c$, where $c \in (-1, 2)$ such that

$$f'(c) = \frac{f(2) - f(-1)}{2 - (-1)}$$

$$3c^2 - 2c - 1 = \frac{3 - 0}{3}$$

$$3c^2 - 2c - 1 = 1$$

$$3c^2 - 2c - 2 = 0$$

$x = 1.08$ or $x = -0.412$

$x = 1.22$ or

$x = -0.55$

prove at least one root exists. ① only one root

Example 4.9. Use the Intermediate Value Theorem and the Mean Value Theorem to prove that the equation $x^3 + x - 1 = 0$ has exactly one real root .

△ 前提：continuity

Let $f(x) = x^3 + x - 1$. We know that $f(x)$ is a polynomial function which is continuous and exists $\forall x \in \mathbb{R}$

Using IVT

$f(0) = -1$

$f(1) = 1$

① According to IVT, there exists at lease one $x = c$, where $c \in (0,1)$. such that $f(c) = 0$

Using MVT

meaning $f(a)$ and $f(b)$ are 0

Assume that there are 2 roots, saying a and b, where $a < b$.

According MVT, there should exist a $x = c$, such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$3x^2 + 1 = 0$$

$$3c^2 + 1 = 0$$

$$c^2 = -\frac{1}{3} \quad \text{(impossible)}$$

∴ $f(x)$ has one and only one root.

Coordinate with MVT, the assumption of 2 roots is wrong.

Example 4.10. Let $f(x) = x^{\frac{2}{3}}$ on $[-8, 27]$. Show that the conclusion of the Mean Value Theorem fails and figure out why.

where $c \in [-8, 27]$

Coordination of MVT tells us there exists a $x = c$, such that

$$f'(c) = \frac{f(27) - f(-8)}{27 - (-8)}$$

$$\frac{2}{3} \cdot c^{-\frac{1}{3}} = \frac{27^{\frac{2}{3}} - (-8)^{\frac{2}{3}}}{27 + 8}$$

$$c = \left(7 \times \frac{2}{3}\right)^2 \approx 101.629$$

MVT failed since c is not between -8 and 27

This is because our function is not differentiable at $x = 0$

$\sqrt[3]{x^2}$

Homework

Example 4.11. Suppose that an object has position $s(t) = t^2 - t - 2$. Find the average velocity over the interval $[3, 6]$ and find the time at which the instantaneous velocity equals the average velocity.

velocity: $v = s'(t) = 2t - 1$

Average V: $v_{avg} = \frac{s(6) - s(3)}{6 - 3} = 8$

$v(t) = 2t - 1 = 8$

$t = 4.5$

Homework

Turning points

Example 4.12. If $f(0) = -3$ and $f'(x) \leq 5$ for all x, how large can $f(2)$ possibly be?

As $f(x)$ exists for all x.

By MVT, we have

$$f(2) - f(0) = f'(c)(2 - 0)$$

$$f(2) = 2f'(c) - 3$$

$\because f'(x) \leq 5$

$\therefore$ the largest $f(2) = 2 \times 5 - 3 = 7$

- **Theorem**

rate of change ∵ F(x) 不变

If $F'(x) = 0$ for all x in (a, b), then $F(x)$ is a constant C on (a, b).

If $F'(x) = G'(x)$ for all x in (a, b), then there is a constant C such that

意味着 2 line are parallel (graph)

$$F(x) = G(x) + C$$ 两线平行

for all x in (a, b).

Example 4.13. Prove that $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$. C 初次解决

F(x) - (-G(x))

Let $f(x) = \sin^{-1} x \Rightarrow f'(x) = \frac{1}{\sqrt{1-x^2}}$

$g(x) = \cos^{-1} x \Rightarrow g'(x) = \frac{-1}{\sqrt{1-x^2}}$

let $F(x) = f(x)$

$G(x) = -g(x)$

$\Rightarrow F'(x) = G'(x)$

According to the theorem

$F(x) = G(x) + C$

$\sin^{-1} x = -\cos^{-1} x + C$

$\sin^{-1} x + \cos^{-1} x = C$

let $x = 0$, $\sin^{-1}(0) + \cos^{-1}(0) = \frac{\pi}{2}$

$\sin^{-1}(x) + \cos^{-1}(x) = \frac{\pi}{2}$ Show

Domain $\sin^{-1}(x)$ and $\cos^{-1}(x)$ are $[-1, 1]$

Exercises 4.2 1, 3, 7, 9, 13, 17, 19, 21, 25, 29, 33.

4.3 Monotonicity and Concavity

- **Monotonicity Theorem**

> Let f be continuous on an interval I and differentiable at every interior point of I.
>
> (i) If $f'(x) > 0$ for all x interior to I, then f is **increasing** on I.
>
> (ii) If $f'(x) < 0$ for all x interior to I, then f is **decreasing** on I.

Example 4.14. If $f(x) = 2x^3 - 3x^2 - 12x + 7$, find the intervals on which f is increasing or decreasing.

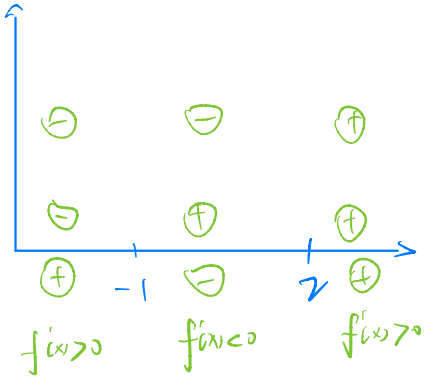
$f'(x) = 6x^2 - 6x - 12$

stationary point

let $f'(x) = 6x^2 - 6x - 12 = 0$

$6(x-2)(x+1) = 0$

$x = 2$ or $x = -1$





$f'(x) > 0$ when $x < -1$ or $x > 2$

$f(x)$ is increasing when $x < -1$ or $x > 2$

$f(x) < 0$ when $-1 < x < 2$

$f(x)$ is decreasing when $-1 < x < 2$

Homework

Example 4.15. On what intervals is $g(x) = \dfrac{x}{1+x^2}$ increasing? On what intervals is it decreasing?

$g'(x) = \dfrac{1+x^2 - x \cdot 2x}{(1+x^2)^2}$

$= \dfrac{1-x^2}{(1+x^2)^2}$

$\because (1+x^2)^2 > 0$ for all x

let $h(x) = 1 - x^2 = 0$

$x = \pm 1$

$1 - x^2$ | $-$ | $+$ | $-$

-1 $\quad$ 1

$\therefore g'(x) < 0$ when $x < -1$ or $x > 1$

$\therefore g(x)$ is decreasing when $x < -1$ or $x > 1$

$g'(x) > 0$ when $-1 < x < 1$

$\therefore g(x)$ is increasing when $-1 < x < 1$

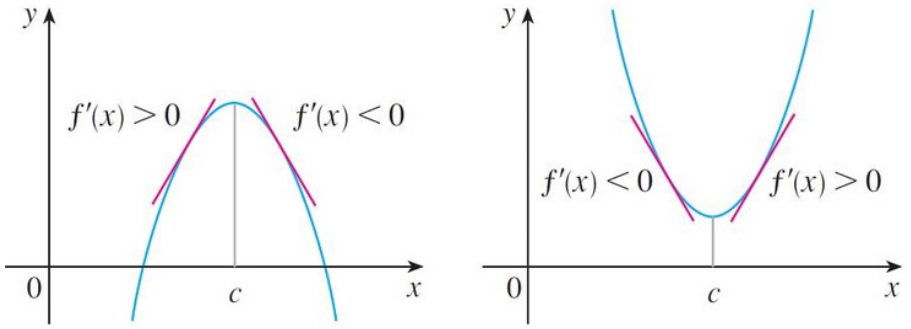
- **Local Extreme Values – First Derivative Test**

Let f be continuous on an open interval (a, b) that contains a critical point c.

(i) If $f'(x) > 0$ for all x in (a, c) and $f'(x) < 0$ for all x in (c, b), then $f(c)$ is a **local maximum** value of f.

(ii) If $f'(x) < 0$ for all x in (a, c) and $f'(x) > 0$ for all x in (c, b), then $f(c)$ is a **local minimum** value of f.

(iii) If $f'(x)$ has the same sign on both sides of c, then $f(c)$ is not a local extreme value of f.



Example 4.16. Find the local extreme values of $f(x) = x^3 - 3x^2 - 9x + 12$ on $(-\infty, \infty)$ using the First Derivative Test.

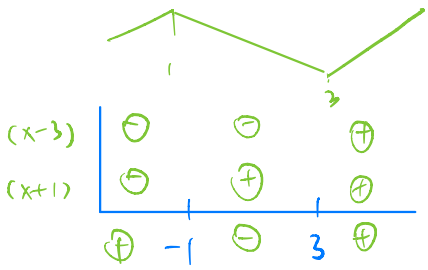
① stationary point

$f'(x) = 3x^2 - 6x - 9 = 0$

$3(x^2 - 2x - 3) = 0$

$3(x-3)(x+1) = 0$

$x = 3$ or $x = -1$



$f'(x) > 0$ when $x < -1$ or $x > 3$

$f(x)$ is increasing ...

$f'(x) < 0$ when $-1 < x < 3$

$f(x)$ is decreasing ...

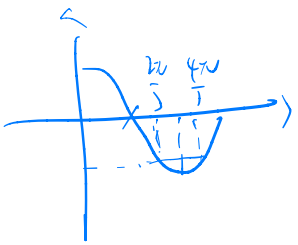
$\therefore f(x)$ has local minimum at $x = 3$

② No Singular point

local maximum at $x = -1$

Homework

Example 4.17. Find the local extreme values of $f(x) = x + 2\sin x$ on $0 \le x \le 2\pi$ using the First Derivative Test.



$f'(x) = 1 + 2\cos x$

Stationary point:

$f'(x) = 1 + 2\cos x = 0$

$\cos x = -\frac{1}{2}$

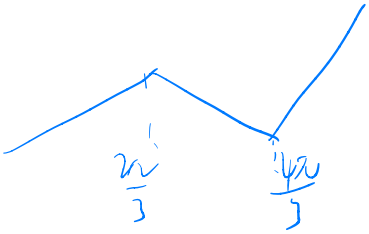
$x = \frac{2\pi}{3}$ or $x = \frac{4\pi}{3}$

$\therefore$ $f'(x) > 0$ when $0 \le x < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < x \le 2\pi$

$f(x)$ is increasing when —

$f'(x) < 0$ when $\frac{2\pi}{3} < x < \frac{4\pi}{3}$

$f(x)$ is decreasing when —



$\therefore$ $f(x)$ has a local minimum at $x = \frac{4\pi}{3}$

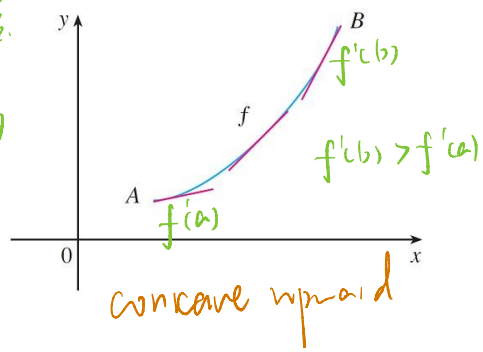
$f(\frac{4\pi}{3}) = \frac{4\pi}{3} + 2\sin(\frac{4\pi}{3}) = 2.45$

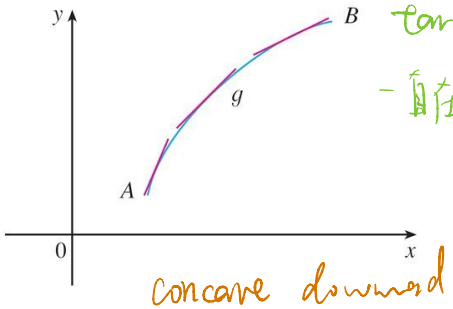
$f(x)$ has a local maximum at $x = \frac{2\pi}{3}$

$f(\frac{2\pi}{3}) = \frac{2\pi}{3} + 2\sin(\frac{2\pi}{3}) = 3.83$

- **Definition**

If the graph of f lies above all its tangent lines on an interval I, then we say that f is **concave upward** on I. If the graph of f lies below all its tangent lines on an interval I, then we say that f is **concave downward** on I.





Let f be differentiable on an open interval I. We say that f is **concave up** on I if f' is increasing on I, and we say that f is **concave down** on I if f' is decreasing on I.

- **Concavity Theorem**

Let f be twice differentiable on an open interval I.

(a) If $f''(x) > 0$ for all x in I, then f is **concave up** on I.

(b) If $f''(x) < 0$ for all x in I, then f is **concave down** on I.

Example 4.18. On what intervals is $f(x) = x^3 - 3x^2 - 9x + 12$ concave upward or concave downward?

$f''(x)$



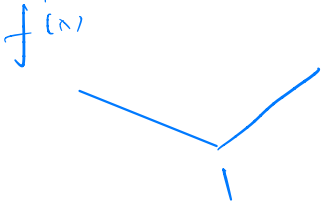
$f'(x) = 3x^2 - 6x - 9$

$f''(x) = 6x - 6$

Let $f''(x) = 6x - 6 = 0$

$x - 1 = 0$

$x = 1$



when $x < 1$, $f''(x) < 0$ $\Rightarrow$ $f(x)$ is concave downward

$x > 1$, $f''(x) > 0$ $\Rightarrow$ $f(x)$ is concave upward

Homework

Example 4.19. Given the function $g(x) = \dfrac{x}{1+x^2}$. Find the intervals of concavity.

$g'(x) = \dfrac{(1+x^2) - x \cdot 2x}{(1+x^2)^2} = \dfrac{1-x^2}{(1+x^2)^2}$

$(1+2x^2+x^4)(-2x) - 4x(1+x^2)(1-x^2)$

$g''(x) = \dfrac{(1+x^2)^2(-2x) - 2x(1-x^2)}{(1+x^2)^4}$

$\dfrac{(1+x^2)^2(-2x) - 2(1+x^2)2x \cdot (1-x^2)}{(1+x^2)^4}$

$= \dfrac{-4x}{(1+x^2)^2}$ $= \dfrac{2x^3 - 4x^3 - 6x}{(1+x^2)^4}$ $= \dfrac{2x(x^2+1)(x^2-3)}{(1+x^2)^4}$ $= \dfrac{2x(x^2-3)}{(1+x^2)^3}$

$\because (1+x^2)^2 > 0$ for all x

$\therefore$ when $x > 0$, $-4x < 0$, $g''(x) < 0$, $g(x)$ concave downward.

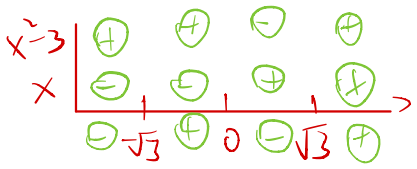
$x < 0$, $-4x > 0$, $g''(x) > 0$, $\cdots$ upward

let $h(x) = 2x(x^2-3) = 0$

$x = 0$ or $x = \pm\sqrt{3}$

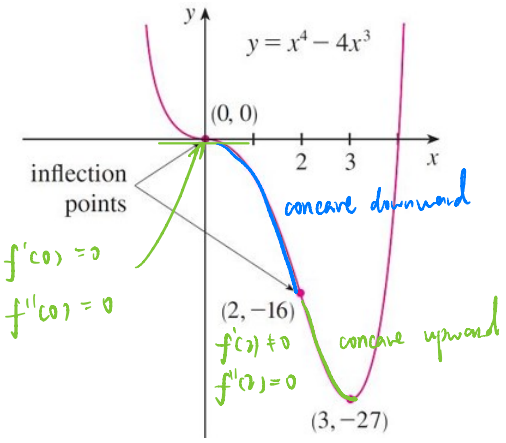
when $x < -\sqrt{3}$ or $0 < x < \sqrt{3}$, $f'' < 0$, concave downward

when $-\sqrt{3} < x < 0$ or $x > \sqrt{3}$, $f'' > 0$, concave upward.



- **Inflection Points**

  We call $(c, f(c))$ an **inflection point** of the graph f if f is concave up on one side of c and concave down on the other side.



- If $(c, f(c))$ is an inflection point, then either $f''(c) = 0$ or $f''(c)$ does not exist.

Example 4.20. Find the inflection points of $F(x) = x^{\frac{1}{3}} + 2$.

$F'(x) = \frac{1}{3} x^{-\frac{2}{3}}$

$F''(x) = -\frac{2}{9} x^{-\frac{5}{3}}$ ($F''(x) = 0$ impossible 也列出)

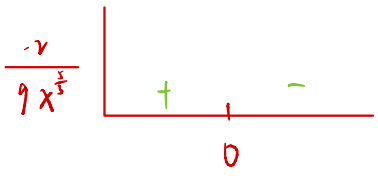
when $F''(x) = -\frac{2}{9x^{\frac{5}{3}}} = 0$ (impossible)

$F''(x)$ DNE when $x = 0$

*

① $\sqrt[3]{x^5}$ 根开方数 x^5 可以是负的 此处可以 apply 到所有的奇数次方根 eg. 3, 5, 7

② 对于偶数次方根，根开方数必须为负



$F''(x) > 0$ when $x < 0$

$F''(x) < 0$ when $x > 0$

} $F(x)$ has an inflection point at $x = 0$

if $f'(c) = 0$ or $f'(c)$ DNE but c is an interior point of domain

$f'(x) < 0$ when $x < c$

$f'(x) > 0$ when $x > c$

$\Rightarrow$ local minimum

$f'(x) > 0$ when $x < c$

$f'(x) < 0$ when $x > c$

$\Rightarrow$ local maximum

$f''(x) < 0$ when $x < c$

$f''(x) > 0$ when $x > c$

$\Rightarrow$ inflection point

$f''(x) > 0$ when $x < c$

$f''(x) < 0$ when $x > c$

$\Rightarrow$ inflection point

- **Local Extreme Values – Second Derivative Test** Not applicable on singular point

> Let f' and f'' exist at every point in an open interval (a, b) containing c, and suppose $f'(c) = 0$. stationary point
>
> (i) If $f''(c) < 0$, then $f(c)$ is a **local maximum** value of f.
>
> (ii) If $f''(c) > 0$, then $f(c)$ is a **local minimum** value of f.

$f''(c) = 0$ inconclusive → use first derivative test

NOTE The Second Derivative Test is inconclusive when $f''(c) = 0$. In other words, at such a point there might be a maximum, there might be a minimum, or there might be neither (as in Example 6). This test also fails when $f''(c)$ does not exist. In such cases the First Derivative Test must be used. In fact, even when both tests apply, the First Derivative Test is often the easier one to use.

Example 4.21. Find (if any) the local maximum and minimum values of $f(x) = x^4 - 4x$ on $(-\infty, \infty)$ using the Second Derivative Test.

$f'(x) = 4x^3 - 4 = 0$ stationary point

$4(x^3 - 1) = 0$

$x = 1$

直接代入 $f'(x) = 0$ 即 $x = 1$

$f''(x) = 12x^2 = 0$

$x = 0$

$f''(1) = 12 > 0$ (local min)

开口向上

$f''(x) > 0$ when $-\infty < x < 0$ or $0 < x < \infty$

minimum

∴ when $x = 1$, there is a local maximum value

$f(1) = 1 - 4 = -3$

Example 4.22. Find (if any) the local maximum and minimum values of

$$G(p) = \frac{1}{p(1-p)}$$

on $(0, 1)$ using the Second Derivative Test.

$$G'(p) = \frac{-(1-p)+p}{p^2(1-p)^2} = \frac{-1+2p}{p^2(1-p)^2}$$

$$G'(p) = \frac{-1+2p}{p^2(1-p)^2} = 0$$

$$-1+2p = 0$$

$$p = \frac{1}{2}$$

$$G''(p) = \frac{p^2(1-p)^2 \cdot 2 - (-1+2p)[2p(1-p)^2 - 2p^2(1-p)]}{p^4(1-p)^4}$$

$$(-1+2p)(1-p)[2p(1-p) - 2p^2)$$

$$f'(x) = \frac{4 - x}{x^{1/3}(6 - x)^{2/3}} \qquad f''(x) = \frac{-8}{x^{4/3}(6 - x)^{5/3}}$$

Example 4.23. Given the function $f(x) = x^{\frac{2}{3}}(6 - x)^{\frac{1}{3}}$.

(a) Find the intervals on which f is increasing or decreasing.

$f'(x) = \frac{2}{3}x^{-\frac{1}{3}}(6-x)^{\frac{1}{3}} + x^{\frac{2}{3}} \cdot \frac{1}{3}(6-x)^{-\frac{2}{3}}(-1)$

$= \frac{2(6-x)^{\frac{1}{3}}(6-x)^{\frac{2}{3}}}{3x^{\frac{1}{3}}(6-x)^{\frac{2}{3}}} - \frac{x^{\frac{2}{3}} \cdot x^{\frac{1}{3}}}{3(6-x)^{\frac{2}{3}} \cdot x^{\frac{1}{3}}}$

$= \frac{2(6-x) - x}{3(6-x)^{\frac{2}{3}}x^{\frac{1}{3}}}$

$= \frac{12-3x}{3(6-x)^{\frac{2}{3}}x^{\frac{1}{3}}} \Rightarrow = \frac{4-x}{x^{\frac{1}{3}}(6-x)^{\frac{2}{3}}}$

$f'(x) = 0$ when $x = 4$

$f'(x)$ DNE when $x = 0$ or $x = 6$

	$x<0$	$0<x<4$	$4<x<6$	$x>6$
$12-3x$	+	+	−	−
$(6-x)^{\frac{2}{3}}$	+	+	+	+
$x^{\frac{1}{3}}$	−	+	+	+
	⊖ 0	⊕ 4	⊖ 6	⊖

$f(x)$ is increasing when $0 < x < 4$ ($f'(x)$ DNE at $x = 6$)

$f(x)$ is decreasing when $x < 0$ or $x > 4$ ($4 < x < 6$ or $x > 6$)

(b) Find the local maximum and minimum values of f.

from a, $f(x)$ has a local minimum value of $f(0)$

$f(0) = 0 \cdot (6-0)^{\frac{1}{3}} = 0$

maximum value of $f(4)$

$f(4) = 4^{\frac{2}{3}}(6-4)^{\frac{1}{3}}$

$= 5.1748$



(c) Find the intervals of concavity and the inflection points. $\Rightarrow f''(x)$

$f'(x) = \frac{12-3x}{3(6-x)^{\frac{2}{3}}x^{\frac{1}{3}}}$

分子是 $(uv)'$

$= uv' + u'v$

$f''(x) = \frac{-3 \times 3(6-x)^{\frac{1}{3}}x^{\frac{1}{3}} - 3x\frac{1}{3}(6-x)^{-\frac{2}{3}}(-1) \cdot \frac{1}{3}x^{-\frac{2}{3}}}{9(6-x)^{\frac{2}{3}}x^{\frac{2}{3}}}$ $\quad 12-3x\left[3(6-x)^{\frac{2}{3}} \cdot \frac{1}{3}x^{-\frac{2}{3}} + x^{\frac{1}{3}} \cdot \frac{2}{3}(6-x)^{-\frac{1}{3}}(-1)\right]$

$= f'(x) = \frac{4-x}{x^{\frac{1}{3}}(6-x)^{\frac{2}{3}}}$

Example 4.24. Given the function $y = e^{\frac{1}{x}}$.

(a) Find the intervals on which f is increasing or decreasing.

(b) Find the local maximum and minimum values of f.

(c) Find the intervals of concavity and the inflection points.

Exercises 4.3 1, 9, 13, 15, 17, 21, 37(a)(b)(c), 43(a)(b)(c), 45(a)(b)(c), 47(a)(b)(c), 75, 83.

4.4 Indeterminate Forms and L'Hôpital's Rule

- **L'Hôpital's Rule** Division form of Indeterminant

Suppose

$$\lim_{x\to a} f(x) = 0 \quad \text{and} \quad \lim_{x\to a} g(x) = 0$$

or

$$\lim_{x\to a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x\to a} g(x) = \pm\infty$$

then

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}$$

0/0 또는 ∞/∞ 일때 f(x) 와 g(x) 를 각각 미분. (!!! 곱 quotient rule)

if the limit on the right hand side exists.

transform into these form

Indeterminate forms: $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0\cdot\infty$, $\infty-\infty$, 0^0, ∞^0, 1^∞.

Example 4.25. Use L'Hôpital's rule to evaluate $\lim_{x\to 2}\frac{x-2}{x^2-4}$ and $\lim_{x\to 0}\frac{\sin x}{x}$.

factorization

squeeze theorem

$\lim_{x\to 2}\frac{x-2}{x^2-4} \overset{H}{=} \lim_{x\to 2}\frac{1}{2x} = \frac{1}{4}$

$\lim_{x\to 0}\frac{\sin x}{x} \overset{H}{=} \lim_{x\to 0}\frac{\cos x}{1} = 1$

Example 4.26. Evaluate $\lim_{x\to 0} \dfrac{1-\cos x}{x^2}$.

$\lim_{x\to 0} \frac{1-\cos x}{x^2} \overset{H}{=} \lim_{x\to 0} \frac{\sin x}{2x} \overset{H}{=} \lim_{x\to 0} \frac{\cos x}{2} = \frac{1}{2}$

Example 4.27. Evaluate $\lim_{x\to 1} \dfrac{\ln x}{x-1}$.

$\overset{H}{=} \lim_{x\to 1} \frac{\frac{1}{x}}{1} = 1$

Example 4.28. Evaluate $\lim_{x\to 0} \dfrac{\tan 2x}{\ln(1+x)}$.

$\overset{H}{=} \lim_{x\to 0} \frac{\sec^2 x \cdot 2}{\frac{1}{1+x}}$

$= 2$

Example 4.29. Evaluate $\lim_{x\to 0} \dfrac{\sin x - x}{x^3}$.

$\overset{H}{=} \lim_{x\to 0} \dfrac{\cos x - 1}{3x^2}$

$\overset{H}{=} \lim_{x\to 0} \dfrac{-\sin x}{6x}$

$\overset{H}{=} \lim_{x\to 0} \dfrac{-\cos x}{6}$

$= -\dfrac{1}{6}$

Example 4.30. Evaluate $\lim_{x\to 0} \dfrac{1 - \cos x}{x^2 + 3x}$.

$\overset{H}{=} \lim_{x\to 0} \dfrac{\sin x}{2x+3}$

$= \dfrac{1}{3}$ $\dfrac{0}{3} = 0$ ✓

Example 4.31. Evaluate $\lim_{x\to\infty} \dfrac{e^{-x}}{x^{-1}}$. $\dfrac{0}{0}$

$y = \dfrac{e^{-x}}{x^{-1}}$

$\ln y = \ln e^{-x} - \ln x^{-1}$ (不是式时不能同时取ln)

$\lim_{x\to\infty} \dfrac{e^{-x}}{x^{-1}} \overset{H}{=} \lim_{x\to\infty} \dfrac{e^{-x}(-1)}{-x^{-2}} = \lim_{x\to\infty} \dfrac{e^{-x}}{2x^{-3}}$ × still Indeterminant

---

$\lim_{x\to\infty} \dfrac{e^{-x}}{x^{-1}} = \lim_{x\to\infty} \dfrac{x}{e^x} \overset{H}{=} \lim_{x\to\infty} \dfrac{1}{e^x} = 0$ $(\frac{1}{\infty})$

Example 4.32. Evaluate $\lim_{x\to\infty} \frac{x^{2.5}}{e^x}$. What can you say about $\lim_{x\to\infty} \frac{x^a}{e^x}$ for any positive real number a?

$\lim_{x\to\infty} \frac{x^{2.5}}{e^x} \overset{H}{=} \lim_{x\to\infty} \frac{2.5x^{1.5}}{e^x} \overset{H}{=} \lim_{x\to\infty} \frac{2.5\times1.5x^{1}}{e^x} \overset{H}{=} \lim_{x\to\infty} \frac{2.5\times1.5\times0.5}{\sqrt{x}\cdot e^x} = 0$

For any positive real number a, $\lim_{x\to\infty} \frac{x^0}{e^x}$ will equal to zero

Since with multiple L'Hopital Rule, x^a will eventually become kx^b, where $b<0$

Example 4.33. Show that for any positive real number a, $\lim_{x\to\infty} \frac{\ln x}{x^a} = 0$.

$\lim_{x\to\infty} \frac{\ln x}{x^a} \overset{H}{=} \lim_{x\to\infty} \frac{\frac{1}{x}}{ax^{a-1}} = \lim_{x\to\infty} \frac{0}{ax^{a-1}} = 0$

不管 a 是不是 1 不

if $a=1$, $x^{a-1} = \frac{1}{x^b} = \frac{1}{\infty} = 0$

$= \lim_{x\to\infty} \frac{1}{ax^a} = 0$, since $a>0$

Example 4.34. Find $\lim_{x\to 0^+} \dfrac{\ln x}{\cot x}$.

$\overset{H}{=} \lim_{x\to 0^+} \dfrac{\frac{1}{x}}{-\cot x \csc x} \times -\csc^2 x = \lim_{x\to 0^+} \dfrac{\sin^2 x}{x}$

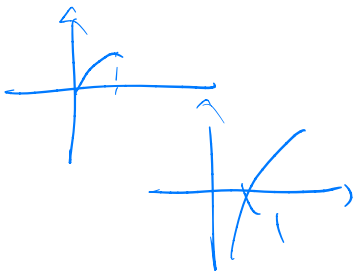
$= \lim_{x\to 0^+} \dfrac{\frac{1}{x}}{-\frac{1}{\tan x \cdot \sin x}}$ $\overset{H}{=} \lim_{x\to 0^+} \dfrac{2\sin x \cos x}{1}$

$= 0$

$= -\lim_{x\to 0^+} \dfrac{\tan x \sin x}{x}$

$\overset{H}{=} -\lim_{x\to 0^+} \dfrac{\sec^2 x \cdot \cos x}{1}$

$= 1$



Example 4.35. Find $\lim_{x\to \frac{\pi}{2}} (\tan x \cdot \ln \sin x)$. ($\infty \cdot 0$)

$\lim_{x\to \frac{\pi}{2}} (\tan x \cdot \ln \sin x) = \lim_{x\to \frac{\pi}{2}} \dfrac{\ln \sin x}{\cot x}$ $\left(\cot x = \dfrac{1}{\tan x}\right)$

$\overset{H}{=} \lim_{x\to \frac{\pi}{2}} \dfrac{\cot x \cdot \frac{\cos x}{\sin x} - \ln \sin x \cdot (-\csc x \cot x)}{\cot^2 x - \csc^2 x}$ ($\csc^2 x$) 分开算.

$\overset{H}{=} \lim_{x\to \frac{\pi}{2}} \dfrac{\frac{\cos x}{\sin x}}{-\csc^2 x}$ $\cdot \sin^2 x$ $(\cot x)' =$

$= \lim_{x}$ $-\frac{\cos}{\sin x}$

$= 0$ $\cot x$

Example 4.36. Find $\lim_{x\to 1^+}\left(\frac{x}{x-1}-\frac{1}{\ln x}\right)$.

$$= \lim_{x\to 1^+} \frac{x\ln x - x + 1}{(x-1)\ln x}$$

$$\overset{H}{=} \lim_{x\to 1^+} \frac{\ln x + 1 - 1}{\ln x + \frac{x-1}{x}}$$

$$\overset{H}{=} \lim_{x\to 1^+} \frac{\frac{1}{x}}{\frac{1}{x}+\frac{1}{x^2}}$$

$$= \frac{1}{2}$$

Example 4.37. Find $\lim_{x\to\infty}(e^x - x)$.

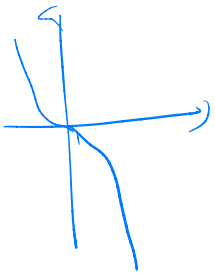
$$= \lim_{x\to\infty} x\left(\frac{e^x}{x} - 1\right)$$

$$\lim_{x\to\infty}\frac{e^x}{x} \overset{H}{=} \lim_{x\to\infty}\frac{e^x}{1} = \infty$$

$$\lim_{x\to\infty} x\left(\frac{e^x}{x} - 1\right) = \infty$$

Example 4.38. Find $\lim_{x\to 0^+}(x+1)^{\cot x}$.

$\frac{1}{\tan x}$



let $y = (x+1)^{\cot x}$

$\ln y = \cot x \ln(x+1)$

$= \frac{\ln(x+1)}{\tan x}$ ($\frac{0}{0}$ indeterminate)

$\lim_{x\to 0^+} \frac{\ln(x+1)}{\tan x} \overset{H}{=} \lim_{x\to 0^+} \frac{\frac{1}{x+1}}{\frac{1}{\cos^2 x}}$

$(\tan x)' = \sec^2 x = \frac{1}{\cos^2 x}$

$= \lim_{x\to 0^+} \frac{\cos^2 x}{x+1} = 0$ | $\ln y = 0$ |

$y = 1$, x $y = e^1 = e$

Example 4.39. Find $\lim_{x\to \frac{\pi}{2}^-}(\tan x)^{\cos x}$.

$\cos x = \frac{1}{\sec x}$

let $y = (\tan x)^{\cos x}$

$\ln y = \cos x \ln(\tan x)$

$\frac{1}{y}\cdot y' = -\sin x \ln(\tan x) + \cos x \cdot \frac{\sec^2 x}{\tan x}$

$\ln y = \frac{\ln(\tan x)}{\sec x}$

$\lim_{x\to \frac{\pi}{2}^-}(\ln y) = \lim_{x\to \frac{\pi}{2}^-}\left[\frac{\ln(\tan x)}{\sec x}\right] \overset{H}{=} \lim_{x\to \frac{\pi}{2}^-} \frac{\frac{\sec^2 x}{\tan x}}{\sec x \tan x}$

$= \lim_{x\to \frac{\pi}{2}^-} \frac{\sec x}{\tan^2 x}$

$= \lim_{x\to \frac{\pi}{2}^-} \frac{\cos x}{\sin^2 x}$

$= 0$

$\lim_{x\to \frac{\pi}{2}^-} y = 1$

Example 4.40. Find $\lim_{x\to 0^+} x^x$.

$\lim_{x\to 0^+} y = 1$

Let $y = x^x$

$\ln y = x \ln x$

$\lim_{x\to 0^+} \ln y = \lim_{x\to 0^+} x \ln x = 0 \cdot (-\infty)$

$= \lim_{x\to 0^+} \frac{\ln x}{\frac{1}{x}}$

$\overset{H}{=} \lim_{x\to 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}}$

$= \lim_{x\to 0^+} \frac{1}{x} \cdot (-x^2)$

$= \lim_{x\to 0^+} (-x) = 0$

Example 4.41. Find $\lim_{x\to 0^+} (\sin x)^{\cot x}$.

Let $y = (\sin x)^{\cot x}$

$\ln y = \cot x \ln(\sin x)$

$\ln y = \frac{\ln(\sin x)}{\tan x}$

$\lim_{x\to 0^+} \ln y = \lim_{x\to 0^+} \frac{\ln(\sin x)}{\tan x} \overset{H}{=} \lim_{x\to 0^+} \frac{\frac{1}{\sin x}}{\sec^2 x} = \lim_{x\to 0^+} \frac{\cos^2 x}{\sin x}$

$\overset{H}{=} \lim_{x\to 0^+} \frac{-2\sin x \cos x}{\cos x}$

$= \lim_{x\to 0^+} \frac{-\sin x}{\cos x} = 0$

$\lim_{x\to 0^+} y = 1 \Rightarrow \lim_{x\to 0^+} (\sin x)^{\cot x} = 1$

Exercises 4.4 7, 11, 13, 15, 17, 19, 21, 23, 27, 31, 33, 35, 39, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 79.

4.5 Summary of Curve Sketching

Step A: State the domain of the function.

- set of values of x which the function is defined.

Step B: Find the x-intercepts and the y-intercepts.

- x-intercepts: set $y = 0$.
- y-intercepts: set $x = 0$.

Step C: Check if the function is even, odd or periodic.

- even: $f(-x) = f(x)$
- odd: $f(-x) = -f(x)$
- periodic: $f(x + p) = f(x)$

Step D: Find the horizontal asymptotes and the vertical asymptotes.

- horizontal:

$$\lim_{x \to -\infty} f(x) = L \quad \text{or} \quad \lim_{x \to \infty} f(x) = L$$

then $y = L$ is the horizontal asymptote.

- vertical:

$$\lim_{x \to a^-} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \to a^+} f(x) = \pm\infty$$

then $x = a$ is the vertical asymptote.

Step E: Find the intervals of increase or decrease.

- find $f'(x)$.

Step F: Find the local maximum and minimum values.

- use First Derivative Test.

Step G: Find the intervals of concavity and the inflection points.

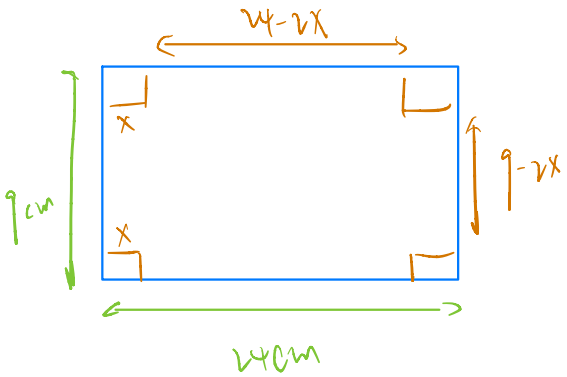
- find $f''(x)$.

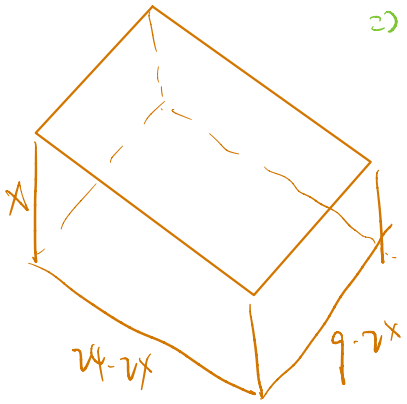
Step H: Sketch the curve.

4.7 Optimization Problems

abs max abs min extreme

Example 4.42. A rectangular box is made from a piece of cardboard 24 cm long and 9 cm wide by cutting out identical squares from the four corners and turning up the sides. Find the dimensions of the box of maximum volume. What is the maximum volume?





$\Rightarrow \begin{cases} 24-2x < 24 \\ 9-2x < 9 \end{cases} \Rightarrow \begin{cases} x < 12 \\ x < 4.5 \end{cases} \Rightarrow 0 < x < 4.5$

Let $V(x)$ be the volume of the box, such that

$$V(x) = (24-2x)(9-2x)x$$

$$= (24-2x)(9x-2x^2)$$

$$= 216x - 48x^2 - 18x^2 + 4x^3$$

$$= 4x^3 - 66x^2 + 216x, \quad 0 < x < 4.5$$

Find its absmax value

No end point.

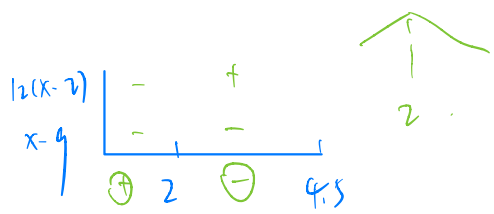
$$V'(x) = 12x^2 - 132x + 216$$

① Stationary point

$$V'(x) = 12x^2 - 132x + 216 = 0$$

$$x = 9 \text{ or } x = 2$$

(reject)



$V(x)$ has a maximum at $x = 2$

Since ...

4.7 Optimization Problems

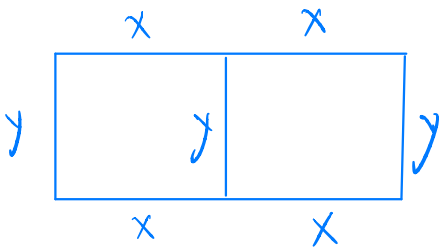
Example 4.42. A rectangular box is made from a piece of cardboard 24 cm long and 9 cm wide by cutting out identical squares from the four corners and turning up the sides. Find the dimensions of the box of maximum volume. What is the maximum volume?

The dimension of the box are 20 cm x 5 cm x 2 cm

The volume is 200 cm^3

Example 4.43. A farmer has 100 meters of wire fence with which he plans to build two identical adjacent pens. What are the dimensions of the enclosure that has maximum area?

△ 圍柵



$4x + 3y = 100$

$3y = 100 - 4x$

$y = \frac{1}{3}(100 - 4x) \quad (0 < x < 25)$

$S = 2xy$

$= \frac{2}{3}(100 - 4x)x$ (a b)

$[\frac{2}{3}ab]' = \frac{2}{3}a'b + \frac{2}{3}ab'$

$\frac{2}{3}(100x - 4x^2)$

$S' = \frac{2}{3}(100 - 4x) + \frac{2}{3}x(-4)$

$= \frac{200}{3} - \frac{8}{3}x - \frac{8}{3}x = \frac{200}{3} - \frac{16}{3}x$

$= 64 - \frac{8}{3}x$

$S' = 0 \Rightarrow x = 12.5$

$\frac{8}{3}x = 64$

$x = 24$

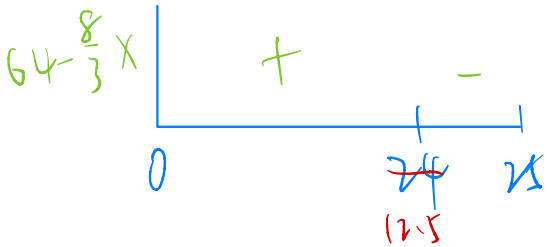
$\lambda \frac{100}{2} = 50$

when $0 < x < 12.5$, $S' > 0$

$S \uparrow$

when $12.5 < x < 25$, $S' < 0$

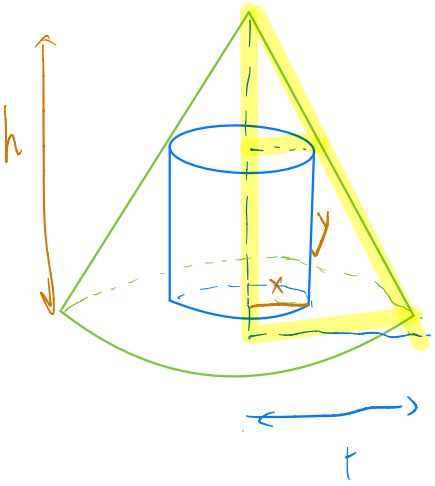
$S \downarrow$



∴ when $x = 12$, the enclosure has maximum area $S(12) = \frac{2}{3}(100 - 4 \times 12)12 = 64\ m^2$

$S(12.5) = \frac{2}{3}(100 - 4 \times 12.5)12.5 = 416.6$

Example 4.44. Find the dimensions of the right circular cylinder of greatest volume that can be inscribed in a given right circular cone.



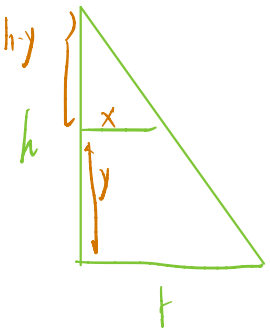
$x \in (0, r)$

Volume of cylinder $= \pi r^2 h$ (general one)

$= \pi x^2 y$

$= \pi x^2 \left(h - \frac{h}{r}x\right)$

$= h\pi x^2 - \frac{h\pi}{r}x^3$



$\frac{h}{r} = \frac{h-y}{x}$

$r(h-y) = hx$

$y = h - \frac{h}{r}x$

$V'(x) = 2xh\pi - \frac{3h\pi}{r}x^2$

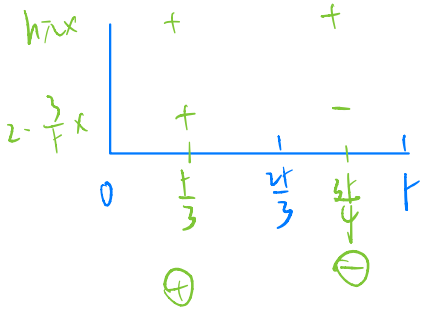
$= xh\pi\left(2 - \frac{3}{r}x\right)$

when $V'(x) = 0$ (find local maximum)

$0 = xh\pi\left(2 - \frac{3}{r}x\right)$

$x = 0$ or $x = \frac{2r}{3}$

因为$\frac{2r}{3}$是此时在区间唯一的正实



when $0 < x < \frac{2r}{3}$, $f'(x) > 0$, $f(x)$ ↑

$\frac{2r}{3} < x < \frac{3r}{4}$, $f'(x) < 0$, $f(x)$ ↓

∴ when $x = \frac{2r}{3}$, $f(x)$ has a maximum value of

$V\left(\frac{2r}{3}\right) = h\pi \times \frac{4r^2}{9} - \frac{h\pi}{r} \cdot \frac{8r^3}{27}$

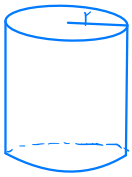
$= \frac{4r^2}{27}h\pi$

Example 4.45. A cylindrical can is to be made to hold 1 L of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can.

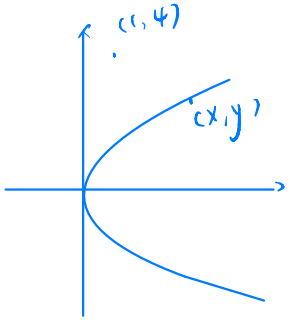
$V = 10^{-3}\ m^3$

$V = \pi r^2 h$

$\text{Area} = 2\pi r^2 + 2\pi r h$



Example 4.46. Find the point on the parabola $y^2 = 2x$ closest to the point $(1, 4)$.



$y^2 = 2x$

$x = \frac{y^2}{2}$

Distance $= \sqrt{(x-1)^2 + (y-4)^2}$

$d(y) = \sqrt{(\frac{y^2}{2} - 1)^2 + (y-4)^2}$

Example 4.47. Find the area of the largest rectangle that can be inscribed in a semicircle of radius r.

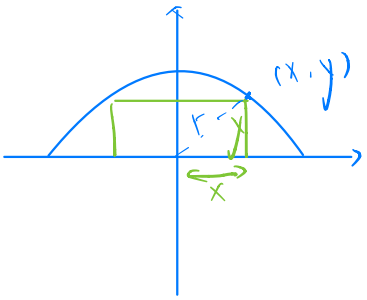
半圓 (take upper half)

要找 $A(x)$

$A'(x)$ → stationary point $A'(x) = 0$

↓

local maximum



$x^2 + y^2 = r^2$

$y = \sqrt{r^2 - x^2}$ $(0 < x < r)$

Area of rectangle $A(x) = 2xy$

$= 2x\sqrt{r^2 - x^2}$

$A'(x) = 2\sqrt{r^2-x^2} + x \cdot \frac{1}{\sqrt{r^2-x^2}} \cdot (-2x)$

$= 2\sqrt{r^2-x^2} - \frac{2x^2}{\sqrt{r^2-x^2}}$

$= \frac{2(r^2-x^2) - 2x^2}{\sqrt{r^2-x^2}} = \frac{-4x^2+2r^2}{\sqrt{r^2-x^2}}$

$A'(x) = 0$

$-4x^2 + 2r^2 = 0$

$x = \frac{r}{\sqrt{2}}$ $(\because x > 0)$

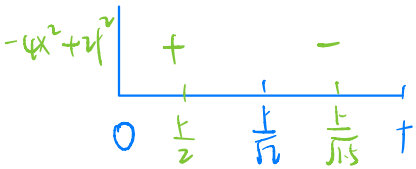
when $0 < x < \frac{r}{\sqrt{2}}$, $A'(x) > 0$, $A(x)$ ↑

$\frac{r}{\sqrt{2}} < x < r$, $A'(x) < 0$, $A(x)$ ↓

∴ when $x = \frac{r}{\sqrt{2}}$, there is a local maximum of

$A\left(\frac{r}{\sqrt{2}}\right) = 2 \cdot \frac{r}{\sqrt{2}}\sqrt{r^2 - \frac{r^2}{2}}$

$= r^2$



Exercises 4.7 3, 5, 7, 13, 15, 23, 26, 31, 37, 73.

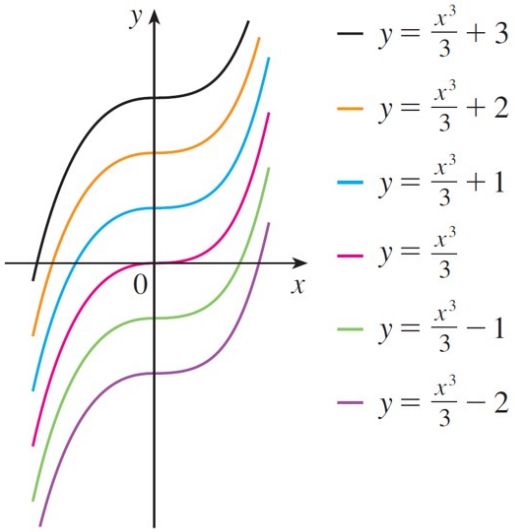
4.9 Antiderivatives

- **Definition** relationship between integral and differential

We call F an **antiderivative** of f on the interval I if $D_x F(x) = f(x)$ on I, that is, if $F'(x) = f(x)$ for all x in I.

Function	Particular antiderivative	Function	Particular antiderivative
$cf(x)$	$cF(x)$	$\sin x$	$-\cos x$
$f(x) + g(x)$	$F(x) + G(x)$	$\sec^2 x$	$\tan x$
$x^n\ (n \neq -1)$	$\dfrac{x^{n+1}}{n+1}$	$\sec x \tan x$	$\sec x$
$\dfrac{1}{x}$	$\ln \lvert x \rvert$	$\dfrac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x$
e^x	e^x	$\dfrac{1}{1+x^2}$	$\tan^{-1} x$
b^x	$\dfrac{b^x}{\ln b}$	$\cosh x$	$\sinh x$
$\cos x$	$\sin x$	$\sinh x$	$\cosh x$

Example 4.48. Find an antiderivative of the function $f(x) = x^2$ on $(-\infty, \infty)$.



If $F(x)$ is an antiderivative of $f(x)$, so is $F(x) + C$ for any constant C.

Example 4.49. Find the antiderivative of $f(x) = 4x^3$ on $(-\infty, \infty)$.

Example 4.50. Find f if $f''(x) = 12x^2 + 6x - 4$ and $f(0) = 4$, $f(1) = 1$.

Example 4.51. A particle moves in a straight line and has acceleration given by $a(t) = 6t + 4$. Its initial velocity is $v(0) = -6$ cm/s and its initial displacement is $s(0) = 9$ cm. Find its position function $s(t)$.

Exercises 4.9 7, 11, 17, 21, 33, 47, 63.